# MA388 Sabermetrics: Lesson 13

Catcher Framing Ability - Visualization

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## Chapter 7

In this chapter, we investigate whether catchers have the ability to consistently frame close pitches, which some believe makes it more likely the umpire will call a strike.

*Do you think some catchers are better than others at framing pitches, and how important is this skill?*

- Video 1
- Video 2

*How could we investigate whether some catchers are better than others? List some challenges with answering this question with data.*

```
library(tidyverse)
library(knitr)
library(baseballr)
```

The code below imports the all pitches thrown in May 2019.

```r
# If you've already collected the pitch data below, just load it.  Otherwise,
# collect it from scratch.

if("statcast_may_2019.rds" %in% dir(getwd())){
  pitches <- readRDS("statcast_may_2019.rds")
}else{

  # Retrieve pitch-level data from May 2019.

get_statcast_pitches <- function(start_day, end_day, chunk_size = 5) {

  # Coerce to Date
  start_day <- as.Date(start_day)
  end_day   <- as.Date(end_day)

  # Create sequence of chunk start dates
  chunk_starts <- seq(start_day, end_day, by = paste(chunk_size, "days"))

  # Build data
  pitch_data <- map_dfr(chunk_starts, function(chunk_start) {

    chunk_end <- min(chunk_start + days(chunk_size - 1), end_day)
       statcast_search(
        start_date = chunk_start,
        end_date   = chunk_end
      )
  })
  return(pitch_data)
}

  # If we want to limit to a certain number of pitches, we might take the head()
  # or sample_n() to get the number we want.
  pitches <- get_statcast_pitches('2019-05-01','2019-05-31')

  saveRDS(pitches, "statcast_may_2019.rds")
}
```

Here is a small sample of the Statcast pitches and variables we've collected.

```
pitches %>%
  select(description, release_speed, pitch_name,
         plate_x, plate_z, balls, strikes, fielder_2) %>%
  head(10) %>%
  kable()
```

| description | release_speed | pitch_name | plate_x | plate_z | balls | strikes | fielder_2 |
|---|---|---|---|---|---|---|---|
| swinging_strike | 79.2 | Changeup | -0.54 | 1.79 | 2 | 2 | 444489 |
| foul | 89.3 | Sinker | -0.74 | 2.42 | 2 | 2 | 444489 |
| foul | 79.5 | Changeup | -0.32 | 1.25 | 2 | 1 | 444489 |
| ball | 88.3 | Sinker | -1.28 | 1.95 | 1 | 1 | 444489 |
| ball | 88.1 | Sinker | -1.82 | 2.29 | 0 | 1 | 444489 |
| called_strike | 87.5 | Sinker | -0.17 | 2.11 | 0 | 0 | 444489 |
| hit_into_play | 78.6 | Changeup | -0.27 | 1.20 | 1 | 2 | 444489 |
| ball | 79.0 | Changeup | -1.25 | 1.23 | 0 | 2 | 444489 |
| foul | 78.5 | Changeup | 0.49 | 1.47 | 0 | 1 | 444489 |
| foul | 86.6 | Sinker | 0.38 | 2.52 | 0 | 0 | 444489 |

One of the variables in `pitches` is the catcher's ID. Let's add the catcher's name to each pitch in our data frame using the master list available at https://www.smartfantasybaseball.com/tools/.

```
# This CSV is available at https://www.smartfantasybaseball.com/tools/.
mlb_IDs <- read_csv("./PLAYERIDMAP.csv")

pitches <- pitches %>%
  left_join(select(mlb_IDs, PLAYERNAME, MLBID),
            by = c("fielder_2" = "MLBID")) %>%
  rename(catcher_name = PLAYERNAME)
```

Let's look at the 10 catchers with the most pitches thrown to them in early May 2019.

```
top10catchers <- pitches %>%
  count(catcher_name) %>%
  arrange(-n) %>%
  head(10) %>%
  pull(catcher_name)

top10catchers
```

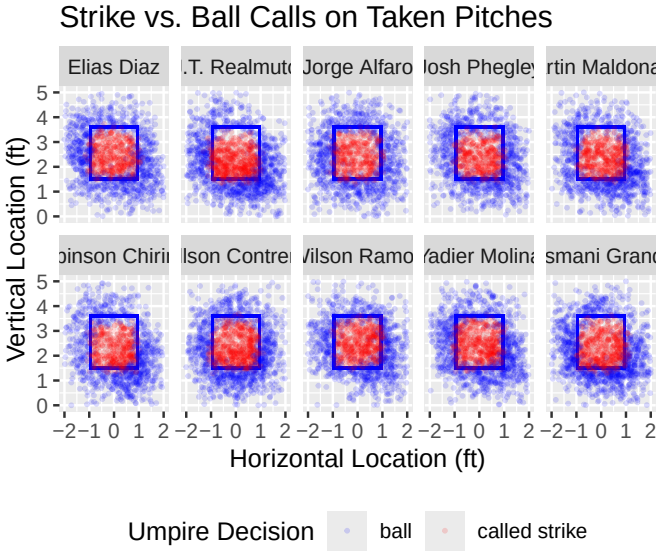
```
 [1] "Yasmani Grandal"   "J.T. Realmuto"     "Willson Contreras"
 [4] "Yadier Molina"     "Elias Diaz"        "Martin Maldonado"
 [7] "Jorge Alfaro"      "Robinson Chirinos" "Wilson Ramos"
[10] "Josh Phegley"
```

Okay, let's see if we can see anything by looking at called strikes versus balls.

```
# Let's plot the strike zone.
# Note px and pz in the book are actually plate_x and plate_z, respectively.
plate_width <- 17 + 2 * (9/pi)
k_zone_plot <- ggplot(NULL, aes(x = plate_x, y = plate_z)) +
  geom_rect(xmin = -(plate_width / 2) / 12,
            xmax = (plate_width / 2) / 12,
            ymin = 1.5,
            ymax = 3.6,
            color = "blue",
            alpha = 0) +
  coord_equal() +
  scale_x_continuous("Horizontal Location (ft)", limits = c(-2,2)) +
  scale_y_continuous("Vertical Location (ft)", limits = c(0,5))

# Look at only taken pitches caught by one of the 10 top catchers.
pitches_taken <- pitches %>%
  filter(catcher_name %in% top10catchers,
         description %in% c("ball", "called_strike")) %>%
  mutate(description = str_replace(description, "_", " "))

# Plot pitches that were balls or called strikes.
k_zone_plot %+% pitches_taken +
  aes(color = description) +
  geom_point(alpha = 0.15, size = .5) +
  scale_color_manual(values = c("blue", "red")) +
  facet_wrap(~catcher_name, ncol = 5) +
  theme(legend.position = "bottom") +
  labs(title = "Strike vs. Ball Calls on Taken Pitches",
       color = "Umpire Decision")
```




How can we make some sense of the different panels in this figure? Look closely – does anything jump out at you?

Okay, this is admittedly pretty subjective, and we can't stop here. We need a quantitative assessment of how different catchers may or may not be contributing to different proportions of called strikes on pitches where batters didn't swing.

```
pitches_taken %>%
  count(catcher_name, description) %>%
  group_by(catcher_name) %>%
  mutate(prop = n / sum(n)) %>%
  filter(description == "called strike") %>%
  arrange(-prop) %>%
  kable(digits = 3,
        caption = "Proportion of taken pitches that were called strikes by catcher.")
```

Table 2: Proportion of taken pitches that were called strikes by catcher.

| catcher_name | description | n | prop |
|---|---|---|---|
| Wilson Ramos | called strike | 504 | 0.362 |
| Willson Contreras | called strike | 575 | 0.356 |
| J.T. Realmuto | called strike | 602 | 0.353 |
| Yadier Molina | called strike | 533 | 0.346 |
| Martin Maldonado | called strike | 496 | 0.337 |
| Robinson Chirinos | called strike | 471 | 0.334 |
| Yasmani Grandal | called strike | 566 | 0.325 |
| Josh Phegley | called strike | 443 | 0.320 |
| Elias Diaz | called strike | 466 | 0.317 |
| Jorge Alfaro | called strike | 427 | 0.302 |

*Does it appear catchers make a difference?*

*Now that we've been talking about this for a while, what new considerations come to mind?*

*How could a model-based approach improve upon this analysis?*